AMC Sprint - Team 1
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Challenge 2 - Verification:

Assessing the impact of verification regime intensity on agreement signature and ratification

Verification Index

To assess the impact of verification intensity / stringency on ratification, we created an **index** out of six existing variables within the amc_vercom_v2 dataset, and assigned them a 'weight' value based on various considerations.

Measure	**Variable**	**Assigned Weight (=1)**
Mechanism_duration	*verified_compliance_mechanism_duration.*	0.125
Trigger_mechanism	*verified_compliance_mechanism _trigger_type, verified_compliance_mechanism_agreement_trigger.*	0.125
Access_level	*Access_level variables weighted (1 - 2) and combined.*	0.25
Phase_access_level	*verified_compliance_mechanism_[phase].*	0.125
Verified_compliance_mechanism	*verified_compliance_mechanism_category.*	0.25
Inspector_type	*verified_compliance_mechanism _inspector_type.*	0.125

Verification, Demonstration, and Consultation

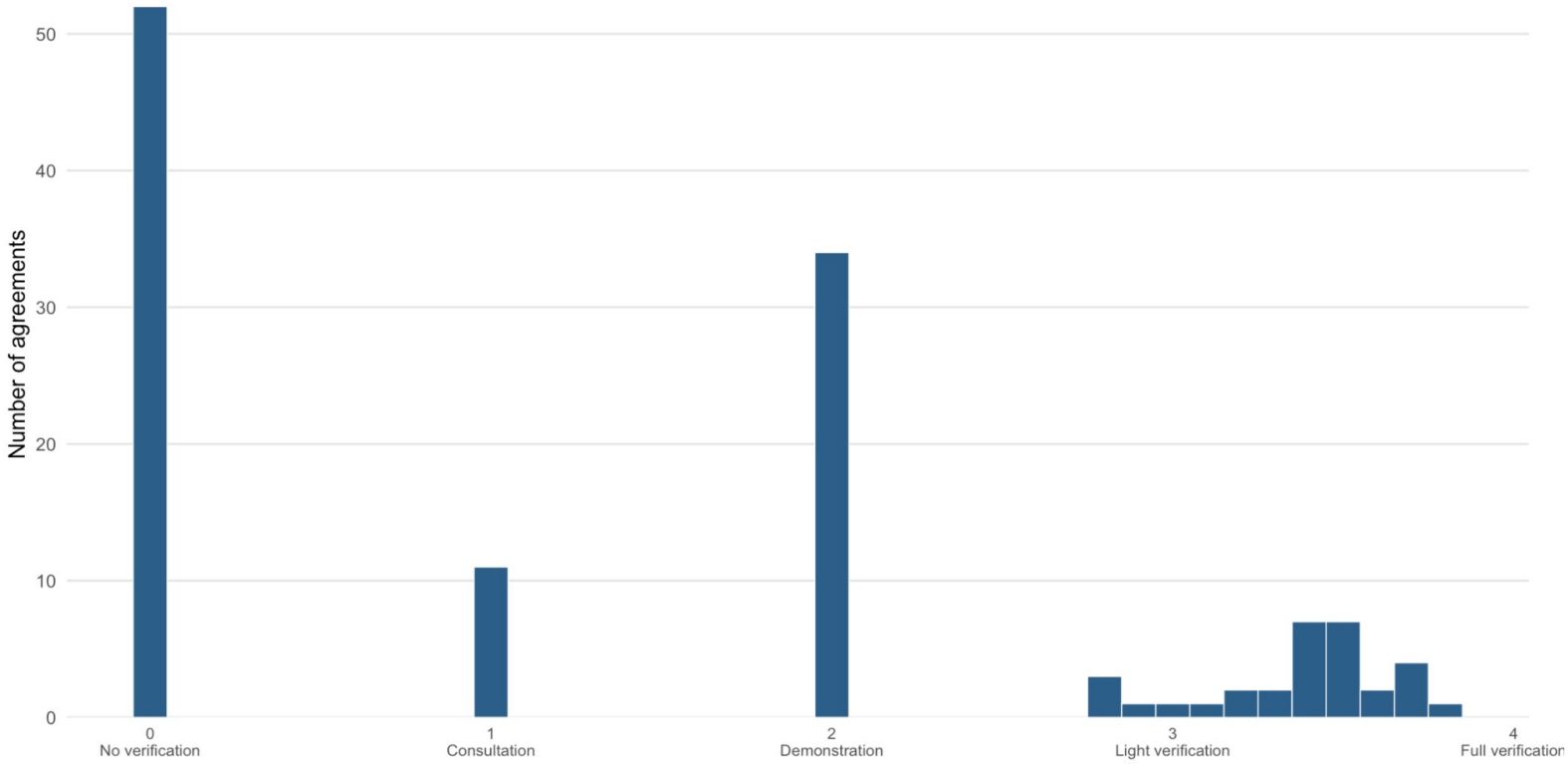
- We found **31 agreements** with **at least 1** verification mechanism
- To construct a measure of verification intensity for the rest of the agreements, we included a rough assessment of the number of demonstration and consultation mechanisms, resulting in an *extended_index*

Extended Verification Intensity Index:

1 - *Consultation mechanism*
2 - *Demonstrated compliance mechanism*
2 - 4 - *Continuum of verification compliance mechanism (low to high)*

Distribution of Verification Intensity by Agreement

Based on extended verification intensity index (0-4)



Source: Alva Myrdal Centre for Nuclear Disarmament /AMC Data

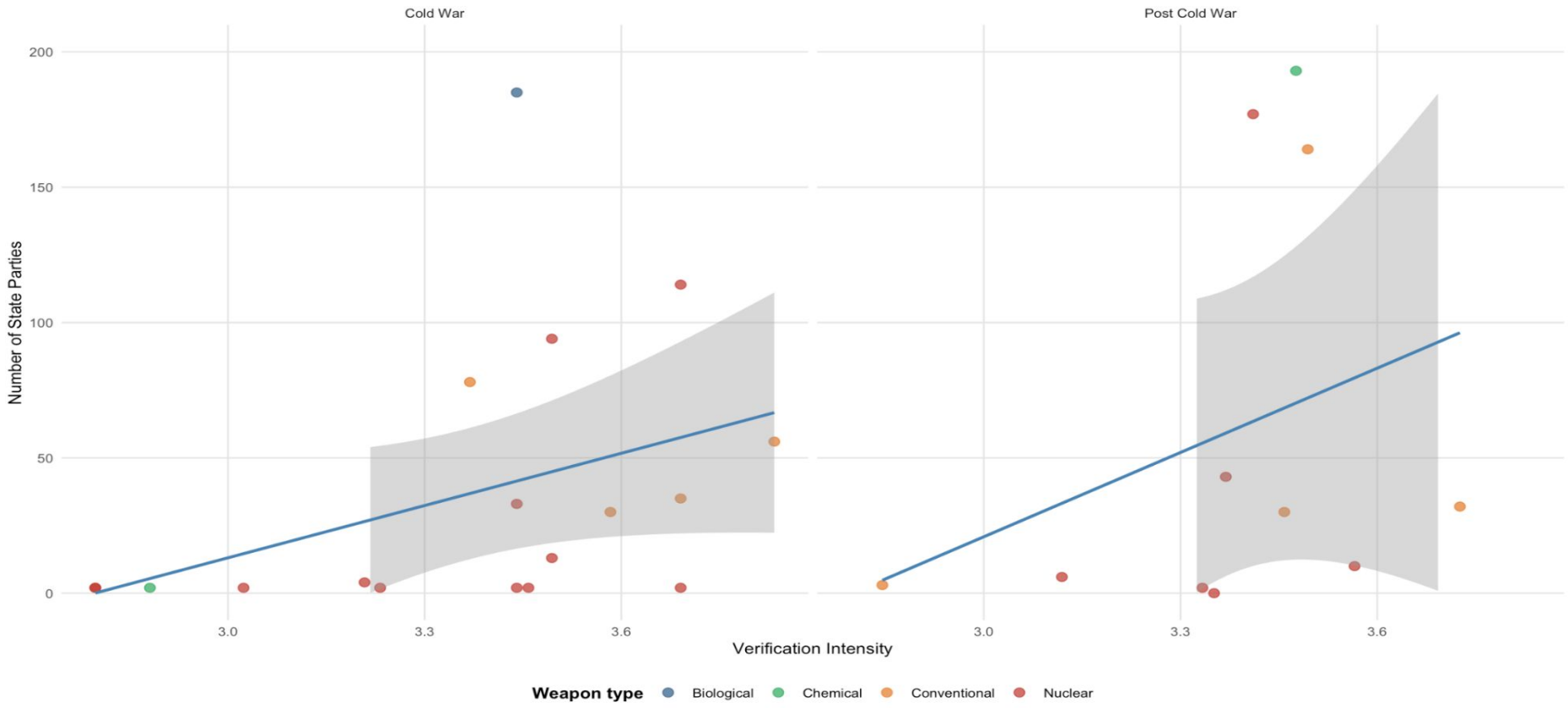
Verification Intensity by Agreement over Time

Each point represents one agreement, colored by weapon type

Source: Alva Myrdal Centre for Nuclear Disarmament /AMC Data

Number of State Parties vs Verification Intensity

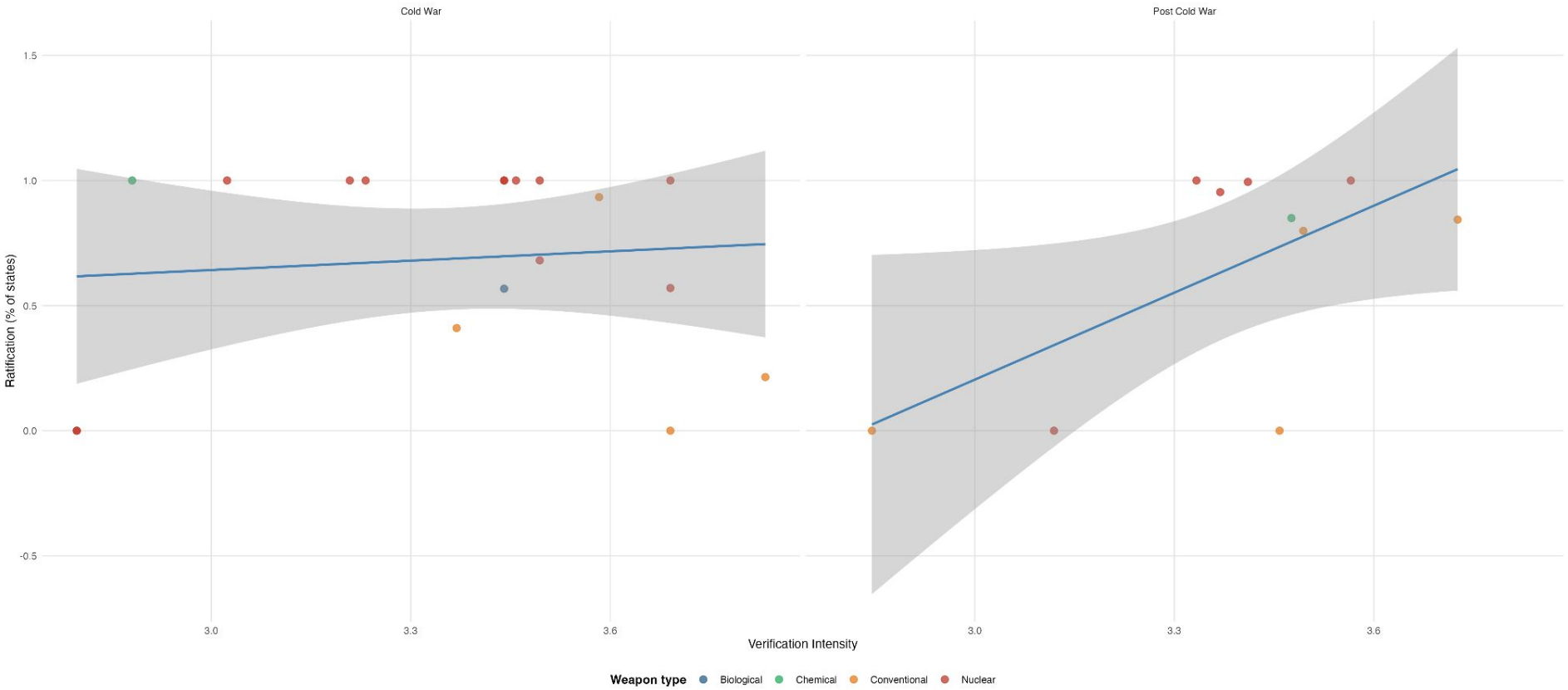
Each point represents one agreement, colored by weapon type



Source: Alva Myrdal Centre for Nuclear Disarmament /AMC Data

Ratification Percentage vs Verification Intensity

Each point represents one agreement, colored by weapon type



Source: Alva Myrdal Centre for Nuclear Disarmament /AMC Data

Controls and Consideration

Addition of control variables to hold constant:

- Weapon type (**weapons_item_topic**)
  - To determine what type of overarching weapon the agreement pertains to (nuclear, chemical, biological, conventional)
- Scope
  - For this control we accumulated various general “bans” and “restrictions” from the amc_weapons_facilities_v2 dataset.
- Era
  - We selected various eras to capture potential deviations in prevailing attitudes, global order, and technology,
  - Eras: Cold War 1945 - 1991;
  - Post Cold War: 1991 - 2001
  - Post 9-11: 2001 - onwards.

Regression findings

Notable difference between the number of states signing/accessing a treaty and the number of states ratifying!

Holding weapon type, era, and scope constant, ***verification stringency makes states more likely to sign an arms control agreement*** (8.5, p<0.01) but it does not have a significant effect on ratification percentage (0.009).

In theory, states may be enticed by the idea of ***confidence-building measures***, which is why we see a ***correlation between higher verification stringency and a higher number of states signing an agreement***.

The impact of verification stringency loses its significance when tested on the proportion of states ratifying agreements.

Verification Intensity and Treaty Participation

	Number of States	Ratification Percentage
(Intercept)	111.149***	-0.023
	(29.065)	(0.320)
extended_verification_intensity	8.598**	0.009
	(2.911)	(0.032)
ban_intensity	0.754	0.129+
	(6.986)	(0.077)
weapon_typeChemical	-96.635**	0.204
	(33.858)	(0.373)
weapon_typeConventional	-115.094***	0.183
	(27.804)	(0.306)
weapon_typeNuclear	-128.487***	0.388
	(28.568)	(0.314)
eraPost 9/11	2.854	0.157
	(11.932)	(0.131)
eraPost Cold War	5.283	0.027
	(8.994)	(0.099)
eraPre Cold War	-34.372*	0.100
	(16.175)	(0.178)
laterality2	40.204***	0.385***
	(8.287)	(0.091)
Num.Obs.	100	100
R2	0.441	0.256
R2 Adj.	0.386	0.182

Future Study

- The current work captures the number of signatories and number of states who ratified, but to accurately measure participation in treaties, it may be useful to measure the **states which *have not* signed agreements for which they are eligible**.

- This would enable a ***measurement of reluctance / refusal to sign / ratify agreements*** that can help test the impact of verification measures.

AI governance

- Verification stringency seems to be a factor that has a varying impact on treaty membership and should be treated with **caution**, especially when considering its differential impact on agreement signature and agreement ratification.

- In nuclear governance, bilateral treaties tend to mitigate this issue and could be a solution for states to have confidence-building measures they feel reassured by. Pertaining to AI governance, this dynamic may be difficult to reproduce given the multipolar and private nature of the AI race.